

Homological Algebra

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Part I

Derived categories

Chapter 1

Derived functors

Chapter 2

Differential graded categories

2.1 Chain complexes

2.1 (Differential graded objects). Let $\mathcal{A}$ be an abelian category. A *complex* over $\mathcal{A}$ is a differential graded object

A differential graded abelian group is traditionally called a *chain complex*.

The following categories are all abelian? projectives and injectives? We have six categories $\text{Ch}_{\geq 0}(\mathcal{A})$ and $\text{Ch}_+(\mathcal{A})$, $\text{Ch}^{\geq 0}(\mathcal{A})$ and $\text{Ch}^+(\mathcal{A})$, $\text{Ch}(\mathcal{A})$ and $\text{Ch}_b(\mathcal{A})$.

(a) $\text{Ch}_{\geq 0}(\mathcal{A})$ is abelian.

Proof. Let $A, B \in \text{Ch}_{\geq 0}(\mathcal{A})$. The cokernel is given as follows. □

2.2 (Differential graded algebra). Let $\mathcal{A}$ be a symmetric monoidal abelian category. A *differential graded algebra* over $\mathcal{A}$ is a monoid object in the category of complexes over $\mathcal{A}$. For a commutative unital ring R , then a differential graded algebra over Mod_R can be characterized by

Let $A \in \mathcal{A}$ be a differential graded algebra. Then, we can consider complexes over A -modules.

derived category of differential graded category.

2.3. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories and suppose $\mathcal{A}$ has enough injectives, that is, every object $A \in \mathcal{A}$ admits a monomorphism $A \rightarrow I$ for an injective object I . Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{B}$ be a left-exact functor.

2.2 Triangulated categories

2.4 (Triangulated categories). A *triangulated category* is an additive functor $\mathcal{D}$ together with a translation functor $\mathcal{D} \rightarrow \mathcal{D} : X \mapsto X[1]$, which is an equivalence of categories, and a collection of distinguished triangles

Chapter 3

Part II

Model categories

Chapter 4

Model structures

4.1

Dashed arrows mean that they exist but may NOT commute with solid arrows in general unless mentioned, and bent dashed arrows mean that they are constructed by the property of projective modules.

model categories are presentations of infinity categories, in the sense that for a model category, the localization at the category of weak equivalences gives rise to an infinity category. For a simplicial model category, homotopy coherent nerve of the category of fibrant-cofibrant objects.

4.1 (Model structures). Following the definition of Hovey, a *model structure* on a category $\mathcal{M}$ is a triple $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ of subcategories of $\mathcal{M}$ whose morphisms are called *weak equivalences*, *cofibrations*, and *fibrations* respectively, and a weak equivalence that is also a cofibration or a fibration is called a *such that*

- (i) (two-out-of-three) $\mathcal{W}$,
- (ii) (retract)
- (iii) (lift) for a commutative square consisting of morphisms f, g, i, p in $\mathcal{M}$ such that $pf = gi$, if i is a cofibration and p are in $\mathcal{C}$ and $\mathcal{F} \cap \mathcal{W}$, or in $\mathcal{C} \cap \mathcal{W}$ and $\mathcal{F}$, respectively, then there exists $h \in \text{Mor}(\mathcal{M})$ such that the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{f} & Y \\ i \downarrow & \nearrow \exists h & \downarrow p \\ X & \xrightarrow{g} & B \end{array} \quad \text{in } \mathcal{M}.$$

- (iv) (factorization)

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \nearrow \exists i & & \nwarrow \exists p \\ & C & \end{array} \quad \text{in } \mathcal{M}$$

A *model category* is a bicomplete category equipped with a model structure.

- (a) opposite category and overcategory
- (b) lifting property and retract closedness
- (c) base change

Proof. To show $\mathcal{C}$ is closed under retracts (in functorial weak factorization system generally), suppose the horizontal composites ACA are the identities, $CD \in \mathcal{C}$, and $XY \in \mathcal{F}$ in the comutative diagram

$$\begin{array}{ccccccc} A & \longrightarrow & C & \longrightarrow & A & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ B & \longrightarrow & D & \longrightarrow & B & \longrightarrow & Y. \end{array}$$

Then, we have the lift DX , and BDX defines a lift BX , so $AB \in \mathcal{C}$. □

4.2. Let $\mathcal{M}$ be a model category. An object X is called *cofibrant* if the unique morphism $0 \rightarrow X$ from the initial object is a cofibration.

The *cofibrant replacement* is the functor $Q : \mathcal{M} \rightarrow \mathcal{M}$ defined by the functorial factorization (if exists) such that $0 \rightarrow QX \xrightarrow{\sim} X$ for all $X \in \mathcal{M}$.

An object Y is called *fibrant* if

For a pointed or unpointed space X , we always have a fibrant replacement $X \rightarrow CX \xrightarrow{\sim} *$

4.3 (Ken Brown lemma). Let $\mathcal{C}$ be a model category. Let $\mathcal{D}$ be a category with weak equivalences. If $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor that takes a acyclic cofibration between cofibrant objects to a weak equivalence, then it takes a weak equivalence between cofibrant objects to a weak equivalence.

Proof. Suppose A and B are cofibrant objects, and AB is a weak equivalence. Factorize $(A \sqcup B)B$ into a cofibration $(A \sqcup B)B' \rightarrow B' \rightarrow B$ followed by a acyclic fibration $B'B$ as we have

$$\begin{array}{ccccc} A & & & & \\ \downarrow & \searrow \sim & & & \\ A \sqcup B & \rightarrow & B' & \rightarrow & B. \\ \uparrow & & & & \\ B & & & & \end{array}$$

Since cofibrations are closed under pushouts, $A(A \sqcup B)$ and $B(A \sqcup B)$ are cofibrations, so are AB' and BB' . On the other hand, AB' and BB' are weak equivalences, and hence are acyclic cofibrations. Then, $F(AB')$ and $F(BB')$ are weak equivalences by the assumption, so $F(AB)$ is a weak equivalence. □

4.4 (Homotopy category of a model category). Let $\mathcal{M}$ be a model category. The *cylinder object* for $X \in \mathcal{M}$ is an object X' such that there is a factorization

$$X \sqcup X \rightarrow X' \xrightarrow{\sim} X.$$

The *path object* for $X \in \mathcal{M}$ is an object Y' such that there is a factorization

$$Y \xrightarrow{\sim} Y' \rightarrow Y \times Y.$$

The functorial cylinder object for X and the path object for Y are denoted by $X \times I$ and Y^I .

left homotopy and right homotopy, cofibrant-fibrant objects.

4.5 (Derived categories and derived functors). For a functor $F : \mathcal{M} \rightarrow \mathcal{N}$ between model categories, a *left derived functor* $LF : \mathbf{h}\mathcal{M} \rightarrow \mathcal{N}$ is defined as the right Kan extension of F with respect to $\mathcal{M} \rightarrow \mathbf{h}\mathcal{N}$. It may not exist in general, but there is an equivalent condition which can be easily investigated.

A sequence $F \rightarrow Y \rightarrow B$ is called a *fiber sequence* if it defines a pullback square

$$\begin{array}{ccc} F & \longrightarrow & Y \\ \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & B \end{array}$$

4.2 Quillen functors

4.3 Examples of model categories

4.6 (Projective model structure on chain complexes). Let $\mathcal{A}$ be an abelian category with enough projective objects and $\text{Ch}_{\geq 0}(\mathcal{A})$ be the category of non-negatively graded chain complexes of $\mathcal{A}$. A *projective model structure* on $\text{Ch}_{\geq 0}(\mathcal{A})$ is defined such that a chain map is

- (i) a weak equivalence if it is a quasi-isomorphism,
- (ii) a cofibration if it is degree-wise monic with the projective cokernel,
- (iii) a fibration if it is degree-wise epic at the strictly positive degrees.

Then, they define a model structure on $\text{Ch}_{\geq 0}(\mathcal{A})$, and its homotopy category becomes the classical derived category.

Proof. Assuming $\text{Ch}_{\geq 0}(\mathcal{A})$ is a bicomplete abelian category, we will check that the projective model structure is indeed a model structure. The two-out-of-three law is clear. We first prove $(\mathcal{C}, \mathcal{F} \cap \mathcal{W})$ is a weak factorization system. Recall that

(lift at degree zero)

Let

$$\begin{array}{ccc} A & \xrightarrow{f} & Y \\ i \downarrow & \sim & \downarrow p \\ X & \xrightarrow{g} & B \end{array} \quad \text{in } \text{Ch}_{\geq 0}(\mathcal{A})$$

be a commutative diagram such that $i : A \rightarrow X$ is a cofibration and $p : Y \rightarrow B$ is a acyclic fibration.

Consider the diagram at the degree zero

$$\begin{array}{ccc} A_0 & \xrightarrow{f_0} & Y_0 \\ i_0 \downarrow & \nearrow \exists t_0 & \downarrow p_0 \\ X_0 & \xrightarrow{g_0} & B_0 \\ \exists s_0 \uparrow & \nwarrow & \downarrow \\ & & P_0 \end{array} \quad \text{in } \mathcal{A},$$

where $i_0 : A_0 \rightarrow X_0$ is monic with projective cokernel P_0 by definition of the cofibration i , and $p_0 : Y_0 \rightarrow B_0$ is epic by the four lemma on the diagram

$$\begin{array}{ccccccc} Y_1 & \xrightarrow{\partial_1} & Y_0 & \longrightarrow & H_0(Y) & \longrightarrow & 0 \\ \downarrow p_1 & & \downarrow p_0 & & \downarrow (p_0)_* & & \\ B_1 & \xrightarrow{\partial_1} & B_0 & \longrightarrow & H_0(B) & \longrightarrow & 0 \end{array} \quad \text{in } \mathcal{A},$$

where every row is exact since the zeroth homology group is the cokernel of the first boundary map, $p_1 : Y_1 \rightarrow B_1$ is epic since p is a fibration, and $(p_0)_* : H_0(Y) \rightarrow H_0(B)$ is an isomorphism since p is a weak equivalence.

By the projectivity of P_0 , we can lift the identity $1 : P_0 \rightarrow P_0$ to get $s_0 : P_0 \rightarrow X_0$ and lift $g_0 s_0 : P_0 \rightarrow B_0$ to get $t_0 : P_0 \rightarrow Y_0$. Since X_0 is the coproduct of A_0 and P_0 , a morphism $h_0 : X_0 \rightarrow Y_0$ is constructed by the universal property. To check h_0 commutes with solid arrows, it suffices to the commutativity with A_0 and B_0 . Then, $h_0 i_0 = f_0$ is clear by definition of h_0 , and $p_0 h_0 = \dots = g_0$.

As for degree zero, we can find $h'_{n+1} : X_{n+1} \rightarrow Y_{n+1}$ such that $h'_{n+1} i_{n+1} = f_{n+1}$ and $p_{n+1} h'_{n+1} = g_{n+1}$. For $n \geq 0$, we also need to check $\partial_{n+1} h'_{n+1} = h_n \partial_{n+1} : X_{n+1} \rightarrow Y_n$, which can fail in general. Define the error $\varepsilon := \partial_{n+1} h'_{n+1} - h_n \partial_{n+1}$.

(factorization)

□

4.7 (Hurewicz model structure on chain complexes).

4.8. Serre and Hurewicz model structures

4.4

monoidal, simplicial, pointed, stable model categories

If we let $*$ be the terminal object of $\mathcal{C}$, then the coslice category or the overcategory $\mathcal{C}_* := * \downarrow \mathcal{C}$, whose objects are morphisms $* \rightarrow B$ in $\mathcal{C}$ for $B \in \mathcal{C}$, is a pointed model category. The addition of disjoint basepoint $\mathcal{C} \rightarrow \mathcal{C}_*$ is left adjoint to the forgetful functor.

Part III

Simplicial categories

Chapter 5

Simplicial sets

5.1 Simplicial sets

5.1 (Simplicial sets). The *simplex category* is the category Δ defined such that objects are totally ordered finite non-empty sets and morphisms are non-decreasing functions. It has a skeleton whose objects are $[n] := \{0, \dots, n\}$ for $n \geq 0$ with the usual order. In Δ , there are special morphisms as in the following diagram

$$[0] \begin{array}{c} \xrightarrow{\delta_1^0} \\ \xleftarrow{\sigma_0^0} \\ \xrightarrow{\delta_1^1} \end{array} [1] \begin{array}{c} \xrightarrow{\delta_2^0} \\ \xleftarrow{\sigma_1^1} \\ \xrightarrow{\delta_2^1} \\ \xleftarrow{\sigma_1^0} \\ \xrightarrow{\delta_2^2} \end{array} [2] \quad \dots$$

defined such that for every $0 \leq i \leq n$

$$\delta_n^i : [n-1] \rightarrow [n] : k \mapsto k + 1_{\geq i}(k), \quad \sigma_n^i : [n+1] \rightarrow [n] : k \mapsto k - 1_{> i}(k),$$

called the *coface maps* and *codegeneracy maps*.

A *simplicial set* is a contravariant functor $X : \Delta^{\text{op}} \rightarrow \text{Set}$. We usually write $X_n := X([n])$ to mean the set of *simplices*, and $d_i^n := X(\delta_n^i) : X_n \rightarrow X_{n-1}$ and $s_i^n := X(\sigma_n^i) : X_n \rightarrow X_{n+1}$ for $0 \leq i \leq n$ are called *face maps* and *degeneracy maps*. It is intuitively something that consists of a sequence X_n of sets that enjoys the combinatorics of face and degeneracy maps. As the image of the skeleton $\{[n]\}$, we have a diagram

$$\dots \quad X_2 \begin{array}{c} \xrightarrow{d_0^2} \\ \xleftarrow{s_1^1} \\ \xrightarrow{d_1^2} \\ \xleftarrow{s_0^1} \\ \xrightarrow{d_2^2} \end{array} X_1 \begin{array}{c} \xrightarrow{d_0^1} \\ \xleftarrow{s_0^0} \\ \xrightarrow{d_1^1} \end{array} X_0$$

A *simplicial map* between simplicial sets is a natural transformation. The category of simplicial sets and simplicial maps is denoted by Set_Δ .

- (a) Every morphism of Δ is uniquely decomposed into the composition of codegeneracy maps on the left and coface maps on the right.

5.2 (Simplicial complexes). A *simplicial complex* is a collection X of non-empty finite subsets of a set V closed under non-empty subsets. We denote by X_n for each $n \geq 0$ the set of all functions $[n] \rightarrow V$ whose image coincides with an element of X . More categorically, it can be essentially defined as a functor $X : \text{Fin}_{>0}^{\text{op}} \rightarrow \text{Set} : [n] \mapsto X_n$ on the category of non-zero finite sets which is concrete in the sense that the canonical map $X_n \rightarrow \prod_{\text{Hom}_\Delta([0],[n])} X_0$ is injective for each $n \geq 0$.

If we chose each skeleton of the simplex category and the category of non-empty finite sets, and if we

denote the objects of each skeleton as $[n]$ for non-negative integer n , then the forgetful functor $\Delta \rightarrow \text{Fin}_{>0}$ gives rise to a linear order on the chosen skeleton of $\text{Fin}_{>0}$, and the composition with the forgetful functor provides a functor $\text{Cpx}_\Delta \rightarrow \text{Set}_\Delta$. We can prove that this is fully faithful.

5.3 (Standard simplices). A *standard simplex* is a simplicial set given by the representable functor in Δ . More precisely, the *standard n -simplex* for $n \geq 0$ is the simplicial set $\Delta^n := \text{Hom}_\Delta(-, [n]) : \Delta^{\text{op}} \rightarrow \text{Set}$. For a simplicial set X , the Yoneda lemma implies $X_n = X([n]) = \text{Hom}_{\text{Set}_\Delta}(\Delta^n, X)$.

The set of homotopy classes of simplicial functions $[X, Y]$?

For $0 \leq i \leq n$, *horns* and *boundaries* are simplicial sets Λ_i^n and $\partial \Delta^n$ defined such that

$$\begin{aligned}\Lambda_i^n([m]) &:= \{\alpha : [m] \rightarrow [n] \mid \alpha([m]) \not\supset [n] \setminus \{i\}\}, \\ \partial \Delta^n([m]) &:= \{\alpha : [m] \rightarrow [n] \mid \alpha([m]) \neq [n]\}.\end{aligned}$$

5.4 (Nerves). We say a simplicial set Y satisfies the *Segal condition* if $Y \rightarrow *$ has the unique right lifting property with respect to the horn inclusions $\Lambda_i^n \rightarrow \Delta^n$ for all $0 < i < n$.

Let $\mathcal{C}$ be a small category. The *nerve* of $\mathcal{C}$ is a simplicial defined as follows.

The nerve, or its geometric realization, of a small category $\mathcal{C}$ is also called the *classifying space* of $\mathcal{C}$. For example, if we interpret a group G as a category of a single object with invertible morphisms, then its nerve gives rise to its classifying space BG .

- (a) The nerve of a small category satisfies the Segal condition.
- (b) A simplicial set satisfying the Segal condition is isomorphic to the nerve of a small category.
- (c) The nerve functor is fully faithful.
- (d) The nerve functor admits a left adjoint, which is called the *homotopy category functor*.

5.5 (Kan complexes). We say a simplicial set Y satisfies the *Kan filler condition* or Y is a *Kan complex* if $Y \rightarrow *$ has the right lifting property with respect to the horn inclusions $\Lambda_i^n \rightarrow \Delta^n$ for all $0 \leq i \leq n$.

- (a) The singular complex of a topological space is a Kan complex.
- (b) A simplicial group is a Kan complex.
- (c) The nerve of a groupoid is a Kan complex.
- (d) The singular complex functor admits a left adjoint, which is called the *geometric realization*.

$$\text{hCW} \rightarrow \text{hCG?} \rightarrow \text{Top}[\mathcal{W}^{-1}]$$

The category of simplicial sets is basically the category of presheaves.

5.2 Model structures on simplicial sets

5.6. bi-completeness, cartesian closed (product and function complex)

5.7 (Kan fibrations).

5.8 (Anodynes).

5.9 (Kan fibrant replacement).

5.10 (Simplicial weak equivalences).

5.11 (Kan-Quillen model structure). Kan and Joyal model structures.

The *Kan-Quillen model structure* on Set_Δ is defined such that a morphism $f : X \rightarrow Y$ of simplicial sets is

- (i) a weak equivalence if the geometric realization $|X| \rightarrow |Y|$ is a homotopy equivalence,
- (ii) a cofibration if it is a monomorphism, i.e. f_n is injective for all $n \geq 0$,
- (iii) a fibration if it is a *Kan fibration*, i.e. it has the right lifting property with respect to the horn pair $\Lambda_i^n \rightarrow \Delta^n$ for all $0 \leq i \leq n$.

5.12 (Joyal model structure).

simplicial homotopy group is defined for Kan complexes? long exact sequences? abelianness for higher?
path object is acyclic cofibration that factors through the diagonal map with a fibration

5.3 Homotopy groups

5.13 (Connected components). For a simplicial set S , $\pi_0(S)$ has many equivalent descriptions... colimit of the diagram $\Delta^{\text{op}} \rightarrow \text{Set}$. left adjoint to the constant inclusion $\text{Set} \rightarrow \text{Set}_\Delta$. coequalizer of two face maps.

Δ^n and $\partial \Delta^n$ are connected...

- (a) If $f : S \rightarrow T$ is a simplicial map and S is connected, then there is a unique component $T' \subset T$ such that $f(S) \subset T'$.
- (b) closed under finite products and π_0 preserves finite products.

Chapter 6

6.1 Simplicial groups

6.2 Simplicial modules

Simplicial methods convert a differential graded category to a simplicial category via the Dold-Kan correspondence, so that a model structure on the differential graded category becomes simplicial. In a simplicial model category we can expand simplicial resolutions explicitly.

6.1 (Dold-Kan correspondence). Let $\mathcal{A}$ be an abelian category. The Dold-Kan correspondence states that the normalized chain complex functor $N : \mathcal{A}_\Delta \rightarrow \text{Ch}_{\geq 0}(\mathcal{A})$ defined such that for $A \in \mathcal{A}_\Delta$ we have $NA_n := \bigcap_{i=1}^n \ker d_i \subset A_n$ and $\partial_n := d_0^n$ is a categorical equivalence. We also have categorical equivalence between the cosimplicial objects $\mathcal{A}^\Delta$ and cochain complexes $\text{Ch}^{\geq 0}(\mathcal{A})$.

It is known that there is a split exact sequence

$$0 \rightarrow NA \rightarrow CA \rightarrow DA \rightarrow 0$$

in $\text{Ch}_{\geq 0}(\mathcal{A})$.

The functor $\Gamma : \text{Ch}_{\geq 0}(\mathcal{A}) \rightarrow \mathcal{A}_\Delta$.

The *Eilenberg-MacLane functor* is the composition $H : \text{Ch}_{\geq 0}(\mathbb{Z}) \xrightarrow{\Gamma} \text{Ab}_\Delta \rightarrow \text{Set}_\Delta$ with the forgetful functor. Note that the cohomology group is computed by the adjoint relations

$$H(X, A) = [N\mathbb{Z}X, A] = [\mathbb{Z}X, \Gamma A] = [X, HA], \quad X \in \text{Set}_\Delta, A \in \text{Ch}_{\geq 0}(\mathbb{Z}).$$

If $A \in \text{Ab}$, then $(HA)_n = K(A, n)$ is the Eilenberg-MacLane space for each $n \geq 0$ so that $H(X, A)$.

Then, the Moore complex functor N is left adjoint to K , i.e. $[NA, C] = [A, KC]$ for $A \in \mathcal{A}_\Delta$ and $C \in \text{Ch}_{\geq 0}(\mathcal{A})$.

Proof. Let $\mathbb{Z}\Delta$ be the pre-additive category generated by the simplex category Δ . □

Note that the homology group is

$$H_n(-, M) : \quad \text{Top} \xrightarrow{\text{Sing}} \text{Set}_\Delta \xrightarrow{(R \cdot) \otimes_R M} (\text{Mod}_R)_\Delta \xrightarrow{C \text{ or } N} \text{Ch}_{\geq 0}(R) \xrightarrow{H_n} \text{Mod}_R$$

6.2. (a) Via the Dold-Kan correspondence $\mathcal{A}_\Delta \cong \text{Ch}_{\geq 0}(\mathcal{A})$, the Kan-Quillen model structure on $\mathcal{A}_\Delta$ induces the projective model structure.

Part IV

Infinity categories

Chapter 7

Models of infinity categories

7.1

All the algebraic or topological structures with finitely many underlying sets will be assumed to be small. For example, groupoids may not be small. Note that Set_Δ is the category of *small* simplicial sets.

For an ordinary category as a nerve, two morphisms are homotopic only if they are identical...

7.1 (Kan complexes). A simplicial set Y is called a *Kan complex* if $Y \rightarrow *$ satisfies the right lifting property with respect to all horns $\Lambda_i^n \rightarrow \Delta^n$. In other words, for every simplicial map $\Lambda_i^n \rightarrow Y$ there is a simplicial map $\Delta^n \rightarrow Y$ such that

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & Y \\ \downarrow & \nearrow & \\ \Delta^n & & \end{array}$$

commutes in the category of simplicial sets for all $n \geq 0$ and $0 \leq i \leq n$.

A geometric model for infinity groupoids. In a Kan complex, including Sing of a topological space, every morphism is invertible up to homotopy.

7.2 (Nerves of categories). A small simplicial set Y is a nerve of a small category if and only if $Y \rightarrow *$ satisfies the right unique lifting property with respect to all inner horns $\Lambda_i^n \rightarrow \Delta^n$. In other words, for every simplicial map $\Lambda_i^n \rightarrow Y$ there is a unique simplicial map $\Delta^n \rightarrow Y$ such that

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & Y \\ \downarrow & \nearrow & \\ \Delta^n & & \end{array}$$

commutes in the category of simplicial sets for all $n \geq 0$ and $0 < i < n$. (Is the smallness of Y important in here?)

7.3 (Weak Kan complexes and quasi-categories). Intuitively, an infinity category is an enriched category over the category of small infinity groupoids, which cannot serve a definition because it includes a circular reasoning.

A possibly large simplicial set Y is called a *infinity category* if $Y \rightarrow *$ has the right lifting property with respect to all horns $\Lambda_i^n \rightarrow \Delta^n$ for $0 < i < n$. We will sometimes call a small infinity category by a *weak Kan complex*.

infinity category, infinity groupoid, infinity bicategory.

Let Top be the category of compactly generated weakly Hausdorff spaces. The singular complex functor

$\text{Sing} : \text{Top} \rightarrow \text{Set}_\Delta$ admits a left adjoint $|\cdot| : \text{Set}_\Delta \rightarrow \text{Top}$ called the geometric realization such that the counit map $|\text{Sing}(X)| \rightarrow X$ is a weak homotopy equivalence.

The *infinity category of spaces*, denoted by Spc , is defined as the homotopy-coherent nerve of the category Grpd_∞ of small infinity groupoids.

7.4. homotopy category, weak homotopy equivalence,

7.2 Fibrations

7.3 Stable infinity categories

examples of stable infinity category: the infinity category of spectra, the dervied category of an abelian category

7.5. A *stable infinity category* is an infinity category such that

- (i) there is a zero object,
- (ii) every morphism admits a fiber and cofiber,
- (iii) a triangle is a fiber sequence if and only if it is a cofiber sequence.

It is known that its homotopy category is triangulated. How can we construct the suspension on the homotopy category?

7.6 (Differential graded categories).

Chapter 8

Infinity topoi